

Problem

Express these sinusoids as phasors:

(a) $v = -14 \sin(5t - 22^\circ) \text{ V}$

(b) $i = -8 \cos(16t + 15^\circ) \text{ A}$

Step-by-step solution

Step 1 of 4

(a)

Write the given sinusoidal expression for voltage.

$$v = -14 \sin(5t - 22^\circ) \text{ V}$$

Consider the following trigonometry property.

$$\cos(A + 90^\circ) = -\sin A$$

Convert the given sine function into cosine function using this property.

$$\begin{aligned} v &= -14 \sin(5t - 22^\circ) \\ &= 14 \cos(5t - 22^\circ + 90^\circ) \\ &= 14 \cos(5t + 68^\circ) \text{ V} \end{aligned}$$

Step 2 of 4

Compare the function 'v' with standard sinusoidal expression $v_m \cos(\omega t + \phi)$.

Here,

The amplitude v_m is 14 V and

The phase angle ϕ is 68° .

Write the phasor domain representation of voltage, v.

$$\begin{aligned} \mathbf{V} &= v_m \angle \phi \\ &= 14 \angle 68^\circ \text{ V} \end{aligned}$$

Therefore, the phasor representation of given sinusoidal function is $14 \angle 68^\circ \text{ V}$.

Step 3 of 4

(b)

Write the given sinusoidal expression for current.

$$i = -8 \cos(16t + 15^\circ) \text{ A}$$

Consider the following trigonometry property.

$$\cos(A - 180^\circ) = -\cos A$$

Convert the given function using this property.

$$\begin{aligned} i &= -8 \cos(16t + 15^\circ) \\ &= 8 \cos(16t + 15^\circ - 180^\circ) \\ &= 8 \cos(16t - 165^\circ) \text{ A} \end{aligned}$$

Step 4 of 4

Compare the function ' i ' with standard sinusoidal expression $i_m \cos(\omega t + \phi)$.

Here,

The amplitude i_m is 8 A and

The phase angle ϕ is -165° .

Write the phasor domain representation of current, i .

$$\begin{aligned} \mathbf{I} &= i_m \angle \phi \\ &= 8 \angle -165^\circ \text{ A} \end{aligned}$$

Therefore, the phasor representation of given sinusoidal function is $8 \angle -165^\circ \text{ A}$.